

Civil Engineering

Calculus and Differential Methods

Revision Booklet

Use these questions for CAT preparation, class discussion and summative revision.

Formula Recall

1. State where the formula below can be used and create one example.

$$d/dx(ax^n) = anx^{n-1}$$

2. State where the formula below can be used and create one example.

$$\int ax^n dx = ax^{n+1}/(n+1) + C$$

3. State where the formula below can be used and create one example.

$$dy/dx + P(x)y = Q(x)$$

4. State where the formula below can be used and create one example.

$$x_{n+1} = x_n - f(x_n)/f'(x_n)$$

5. State where the formula below can be used and create one example.

$$\text{Trapezoidal rule: } \int y dx \approx h/2[y_0 + y_n + 2\sum y_i]$$

Structured Revision Questions

1. A workplace task involves drainage-channel area. Select the correct mathematical method, solve the problem and write a brief interpretation.
2. A workplace task involves settlement-rate models. Select the correct mathematical method, solve the problem and write a brief interpretation.
3. A workplace task involves root finding. Select the correct mathematical method, solve the problem and write a brief interpretation.
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References and Further Reading

These notes are original learning materials prepared for Mr Mwirigi Mathematics Training Hub. They are academically informed by standard engineering mathematics texts and the local TVET curriculum framework. No textbook pages or worked examples have been copied.

1. Stroud, K. A. and Booth, D. J. Engineering Mathematics, 8th edition. Bloomsbury Academic.
2. Bird, J. Engineering Mathematics. Routledge / Taylor & Francis.
3. Bird, J. Higher Engineering Mathematics. Routledge / Taylor & Francis.
4. Relevant Civil Engineering, Building Technology and Land Survey curriculum and occupational standard documents.